

Biochemistry DDS program

1. Vitamin B3 can be synthesized from and its deficiency causes
 - a. Leucine, beriberi
 - b. Methionine, scurvy
 - c. Lysine, pellagra
 - d. Tryptophan, dermatitis
2. Ribitol is found in the structure of which vitamin?
 - a. Thiamine
 - b. Folic acid
 - c. Riboflavin
 - d. Biotin
3. The active form of pantothenic acid is
 - a. NADH
 - b. TPP
 - c. PLP
 - d. Coenzyme A
4. Choose the **wrong statement** about THF.
 - a. It is the active form of folic acid
 - b. It transfers one carbon units
 - c. It can be trapped in folate trap
 - d. It is also called red vitamin
5. Biotin deficiency can be due to
 - a. Lack of intrinsic factor
 - b. Glycoprotein called "avidin"
 - c. Lack of hydroxylation in collagen
 - d. Spina bifida
6. Methyl cobalamin is coenzyme for which enzyme?
 - a. Acetyl coA carboxylase
 - b. Methionine synthase
 - c. Alkaline phosphatase
 - d. Methylmalonyl-CoA mutase



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7. Which is the active form of vitamin D3?
 - a. 25 (OH) D3
 - b. 1, 24 (OH)2 D3
 - c. 1,25 (OH)2 D3
 - d. 7-dehydrocholesterol
8. Vit E is participates in
 - a. Antioxidant activity
 - b. gamma-carboxylation
 - c. proline hydroxylation
 - d. calcium regulation
9. In all enzymes the active site
 - a. Contains the substrate binding site
 - b. Lies in a region of the primary sequence distinct from the substrate binding site
 - c. Contains a metal ion as a prosthetic group
 - d. Contains the amino acid side chains involved in catalyzing the reaction
10. An enzyme can facilitate the rate of a reaction by
 - a. Stabilizing the transition state
 - b. Binding very tightly to the substrate
 - c. Preventing the substrate from changing its ionic state
 - d. Preventing the reaction from proceeding in the reverse direction
11. Turnover number (Kcat)
 - a. Has units of 1/time
 - b. Is inversely related to speed of the reaction
 - c. Is a ratio of the rate constants for the formation of ES and of product
 - d. Has units of substrate concentration
12. When added to a reaction, a catalyst
 - a. Supplies the heats of formation
 - b. Alters the equilibrium constants K_{eq} to favor the formation of products
 - c. Increases the rate at which equilibrium is reached
 - d. Changes the order of the reaction (e.g., first order to second order)



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13. Catalytic efficiency of an enzyme is expressed by
- Kcat
 - Km
 - Kcat/Km
 - Km/Kcat
14. The intercept point of a Lineweaver-Burk Plot with the X-axis is referred to as
- Km
 - 1/Km
 - Vmax
 - 1/Vmax
15. Which one occurs in an uncompetitive inhibition?
- Km increases and Vmax decreases
 - Both Km and Vmax decrease
 - Both Km and Vmax increase
 - Km decreases and V max increases
16. In an enzyme catalyzed reaction, in the presence of and inhibitor, high concentrations of the substrate overcomes the inhibition and V_{max} is reached. What kind of inhibitor is there in the reaction?
- Irreversible
 - Un-competitive
 - Non-competitive
 - Competitive
17. All are true about bond types in DNA except.....
- Covalent bond in the backbone
 - Phosphodiester bonds between bases
 - Hydrogen bonds between complementary base pairs
 - Hydrophobic bonds between successive bases
18. All are true about Z- DNA except
- It is left handed
 - Has about 12 base pair per turn
 - Its backbone is zigzag
 - Is the most abundant type

19. The most different type of RNA in the cell is.....

- a. rRNA
- b. mRNA
- c. tRNA
- d. SnRNA

20. All are true about tRNA except.....

- a. serves as an adaptor molecule during protein synthesis
- b. identify and move amino acids to the site of protein synthesis
- c. Has codon arm to distinguish amino acids and attach to mRNA
- d. carboxyl group of amino acids are linked to it via ester bond

21.not only is the start codon, but also can

- a. Methionine codon – be found in other part of mRNA
- b. AUG – be found in A site at initiation step
- c. AUG – be a synonym codon for few other amino acids
- d. Methionine codon – act as stop codon in translation

22. In Prokaryotes, the ribosomal binding site on mRNA is called?

- a. Hogness sequence
- b. Shine-Dalagarno Sequence
- c. Pribnow sequence
- d. TATA box

23. With respect of the mutation, which statement is correct?

- a. Conversion of AAA to UAA is called nonsense and terminate the polypeptide
- b. Conversion of AAA to AAG is called silent and inactivate the polypeptide
- c. Conversion of AAA to AAG is called missense and terminate the polypeptide
- d. Conversion of AAA to GAG is called nonsense and terminate the polypeptide

24. The enzyme required for transcription is?

- a. Restriction
- b. DNA polymerase
- c. RNA polymerase
- d. RNAase

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25. All are true about enhancer except.....
 - a. They can be downstream from the promoter
 - b. They can be upstream from the promoter
 - c. They can be thousands of base pairs far from the gene it regulates
 - d. They are found in promoter region of the gene of interest

26. All are true about Post-Transcriptional regulation in eukaryotes except.....
 - a. Splice site choice
 - b. Poly A site choice
 - c. mRNA editing
 - d. Promoter selection

27. Which enzyme is incorrectly matched to its function?
 - a. Primase - Transcription
 - b. Topoisomerase - replication
 - c. Translocase - Translation
 - d. Ligase - replication

28. When only lactose is available it attaches to..... and its operon.
 - a. A repressor - activate
 - b. An activator - inactivate
 - c. A repressor - inactivate
 - d. An activator - activate

29. In the absence of, adenyl cyclase is active, and sufficient quantities of cAMP are made and bind to the..... causing RNA polymerase to more efficiently initiate transcription at the promoter region.
 - a. Lactose, repressor
 - b. Glucose, CAP protein
 - c. Allolactose, CAP protein
 - d. Glucose, repressor

30. All are true about regulation of gene expression in eukaryotes, except
 - a. phosphorylation of the eIF-2
 - b. catabolic Operons
 - c. chromatin remodeling
 - d. alternative splicing

31. All enzymes are responsible in repair of UV radiation and Thymine dimer except.....
 - a. Glycosylase
 - b. Endonuclease
 - c. DNA polymerase
 - d. DNA ligase



Short Answer Questions :

32. An enzyme without its coenzyme is called
33. What type of bond is often formed between prosthetic groups and proteins ?
34. The combination of 8 histons rapped by DNA around it, is called.....
35. The binding of an uncharged tRNA to the A site of a ribosome triggers a series of events that leads to the production of
36. Name one antibiotic that prevent transcription.....
37. Which enzyme is responsible for prevention of DNA erosion (shortening) in eukaryotes?.....
38. Production of apoB48 and apoB100 in intestine and liver respectively is because of.....
39. Poly A adenylationthe half-life of mRNA in eukaryotes.
40. In one type of Anemia, called **macrocytic anemia**, abnormally large erythrocytes can be seen in the blood due to the inability of the bone marrow to synthesize DNA during erythrocyte maturation; **what vitamins are used to treat this deficiency symptoms?**